

Ruckig

Phase-Duration Equation Catalog

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1 Scope

This catalog records analytical phase-duration equations used for Ruckig’s third-order, jerk-limited position interface. It covers minimum-duration extremal profiles, fixed-duration time-synchronization profiles, and the piecewise-constant-jerk propagation equations used to check a candidate.

The original Ruckig paper supplies the trajectory-generation framework and profile classification [1]. The implementation-level closed forms cataloged here are transcribed into mathematical notation from the official Ruckig v0.19.4 source: Step 1 for minimum-duration profiles [2], Step 2 for time synchronization [3], and the profile and polynomial-root definitions used for candidate validation and root selection [4, 5].

The catalog uses the implementation’s symmetric physical limits. Direction is handled by applying a sign to the velocity, acceleration, and jerk limits.

2 Notation and propagation

Symbol	Definition
p_0, p_f	Initial and terminal position.
v_0, v_f	Initial and terminal velocity.
a_0, a_f	Initial and terminal acceleration.
$\bar{v}$	Directed velocity limit.
$\bar{a}, \underline{a}$	Directed upper and lower acceleration limits.
J	Directed jerk limit.
Δp	$p_f - p_0$.
Δv	$v_f - v_0$.
Δa	$a_f - a_0$.
T	Requested fixed duration.
$t_1, \dots, t_7$	Seven phase durations.

For a phase with constant jerk j and duration t ,

$$a^+ = a + jt, \tag{1}$$

$$v^+ = v + at + \frac{1}{2}jt^2, \tag{2}$$

$$p^+ = p + vt + \frac{1}{2}at^2 + \frac{1}{6}jt^3. \tag{3}$$

For direction $d \in \{+1, -1\}$,

$$\bar{v} = dv_{\max}, \quad v_{\min}^d = -dv_{\max}, \tag{4}$$

$$\bar{a} = da_{\max}, \quad \underline{a} = -da_{\max}, \tag{5}$$

$$J = dj_{\max}. \tag{6}$$

The minimum-time engine uses the UDDU switching law

$$\mathbf{j}_{\text{UDDU}} = J[1, 0, -1, 0, -1, 0, 1]. \tag{7}$$

The fixed-time engine also uses the UDUD switching law

$$\mathbf{j}_{\text{UDUD}} = J[1, 0, -1, 0, 1, 0, -1]. \quad (8)$$

3 Minimum-time velocity-bound families

3.1 AccelerationVelocity

This family reaches both directed acceleration bounds and the directed velocity bound.

$$t_1 = \frac{-a_0 + \bar{a}}{J}, \quad (9)$$

$$t_2 = \frac{a_0^2/2 - \bar{a}^2 - J(v_0 - \bar{v})}{\bar{a}J}, \quad (10)$$

$$t_3 = \frac{\bar{a}}{J}, \quad (11)$$

$$t_5 = -\frac{\underline{a}}{J}, \quad (12)$$

$$t_6 = -\frac{a_f^2/2 - \underline{a}^2 - J(v_f - \bar{v})}{\underline{a}J}, \quad (13)$$

$$t_7 = t_5 + \frac{a_f}{J}. \quad (14)$$

The velocity-plateau duration is

$$t_4 = \frac{1}{24\bar{a}\underline{a}J^2\bar{v}} \left(3(a_0^4\underline{a} - a_f^4\bar{a}) + 8\bar{a}\underline{a}[a_f^3 - a_0^3 + 3J(a_0v_0 - a_fv_f)] \right) \quad (15)$$

$$+ 6a_0^2\underline{a}(\bar{a}^2 - 2Jv_0) - 6a_f^2\bar{a}(\underline{a}^2 - 2Jv_f) \quad (16)$$

$$- 12J\{\bar{a}\underline{a}[\bar{a}(v_0 + \bar{v}) - \underline{a}(v_f + \bar{v}) - 2J\Delta p] \quad (17)$$

$$+ (\underline{a} - \bar{a})J\bar{v}^2 + J(\bar{a}v_f^2 - \underline{a}v_0^2)\}. \quad (18)$$

3.2 TerminalAccelerationVelocity

Define

$$q_0 = \sqrt{\frac{a_0^2}{2J^2} + \frac{\bar{v} - v_0}{J}}. \quad (19)$$

Then

$$[t_1, t_2, t_3] = [q_0 - a_0/J, 0, q_0], \quad (20)$$

$$t_5 = -\underline{a}/J, \quad (21)$$

$$t_6 = -\frac{a_f^2/2 - \underline{a}^2 - J(v_f - \bar{v})}{\underline{a}J}, \quad (22)$$

$$t_7 = t_5 + a_f/J, \quad (23)$$

and

$$t_4 = -\frac{1}{24\underline{a}J^2\bar{v}} \left(3a_f^4 - 8\underline{a}(a_f^3 - a_0^3) - 24\underline{a}J(a_0v_0 - a_fv_f) \right) \quad (24)$$

$$+ 6a_f^2(\underline{a}^2 - 2Jv_f) - 12J[2\underline{a}J\Delta p + \underline{a}^2(v_f + \bar{v})] \quad (25)$$

$$+ J(\bar{v}^2 - v_f^2) + \underline{a}q_0(a_0^2 - 2J(v_0 + \bar{v}))]. \quad (26)$$

3.3 InitialAccelerationVelocity

Define

$$q_f = \sqrt{\frac{a_f^2}{2J^2} + \frac{\bar{v} - v_f}{J}}. \quad (27)$$

Then

$$t_1 = (-a_0 + \bar{a})/J, \quad (28)$$

$$t_2 = [a_0^2/2 - \bar{a}^2 - J(v_0 - \bar{v})]/(\bar{a}J), \quad (29)$$

$$t_3 = \bar{a}/J, \quad (30)$$

$$[t_5, t_6, t_7] = [q_f, 0, q_f + a_f/J], \quad (31)$$

and

$$t_4 = \frac{1}{24\bar{a}J^2\bar{v}} \left(3a_0^4 + 8\bar{a}(a_f^3 - a_0^3) + 24\bar{a}J(a_0v_0 - a_fv_f) \right) \quad (32)$$

$$+ 6a_0^2(\bar{a}^2 - 2Jv_0) - 12J[-2\bar{a}J\Delta p + \bar{a}^2(v_0 + \bar{v})] \quad (33)$$

$$+ J(\bar{v}^2 - v_0^2) + \bar{a}q_f(-a_f^2 + 2J(v_f + \bar{v})) \Big). \quad (34)$$

3.4 Velocity

With q_0 and q_f as above,

$$[t_1, t_2, t_3] = [q_0 - a_0/J, 0, q_0], \quad (35)$$

$$[t_5, t_6, t_7] = [q_f, 0, q_f + a_f/J], \quad (36)$$

and

$$t_4 = \frac{a_f^3 - a_0^3}{3J^2\bar{v}} + \frac{a_0v_0 - a_fv_f + (a_f^2q_f + a_0^2q_0)/2}{J\bar{v}} \quad (37)$$

$$- \left(\frac{v_0}{\bar{v}} + 1 \right) q_0 - \left(\frac{v_f}{\bar{v}} + 1 \right) q_f + \frac{\Delta p}{\bar{v}}. \quad (38)$$

4 Minimum-time acceleration-bound family

4.1 AccelerationBounds

Let $D = \bar{a} - \underline{a}$ and

$$R = \frac{1}{3DJ^2} \left(3(a_f^4\bar{a} - a_0^4\underline{a}) + \bar{a}\underline{a}[8(a_0^3 - a_f^3) + 3\bar{a}\underline{a}(\bar{a} - \underline{a})] \right) \quad (39)$$

$$+ 6\underline{a}a_f^2 - 6\bar{a}a_0^2 + 12J\{\bar{a}\underline{a}[(\bar{a} - 2a_0)v_0 - (\underline{a} - 2a_f)v_f] \quad (40)$$

$$+ \underline{a}a_0^2v_0 - \bar{a}a_f^2v_f\} + \frac{4(\bar{a}v_f^2 - \underline{a}v_0^2 - 2\bar{a}\underline{a}\Delta p)}{D}. \quad (41)$$

Define

$$r = \frac{1}{2}\sqrt{R}, \quad (42)$$

$$b_0 = \frac{a_0^2}{2\bar{a}J} + \frac{\underline{a} - 2\bar{a}}{2J} - \frac{v_0}{\bar{a}}, \quad (43)$$

$$b_f = -\frac{a_f^2}{2\underline{a}J} - \frac{\bar{a} - 2\underline{a}}{2J} + \frac{v_f}{\underline{a}}. \quad (44)$$

The common phases are

$$[t_1, t_3, t_5, t_7] = [(-a_0 + \bar{a})/J, \bar{a}/J, -\underline{a}/J, (-\underline{a} + a_f)/J], \quad (45)$$

with $t_4 = 0$. Two algebraic branches are evaluated:

$$\text{Branch A: } t_2 = b_0 - r/\bar{a}, \quad t_6 = b_f + r/\underline{a}, \quad (46)$$

$$\text{Branch B: } t_2 = b_0 + r/\bar{a}, \quad t_6 = b_f - r/\underline{a}. \quad (47)$$

5 Minimum-time nonsaturated families

5.1 Unconstrained

Define

$$H = \frac{a_0^2 - a_f^2}{2J} + v_f - v_0. \quad (48)$$

Let τ be an admissible positive real root of

$$0 = \tau^4 - \frac{2[a_0^2 + a_f^2 - 2J(v_0 + v_f)]}{J^2} \tau^2 \quad (49)$$

$$+ \left\{ \frac{4[a_0^3 - a_f^3 + 3J(a_f v_f - a_0 v_0)]}{3J^3} - \frac{4\Delta p}{J} \right\} \tau - \frac{H^2}{J^2}. \quad (50)$$

With $h = H/(2J\tau)$,

$$[t_1, \dots, t_7] = [h + \tau/2 - a_0/J, 0, \tau, 0, 0, 0, -h + \tau/2 + a_f/J]. \quad (51)$$

5.2 InitialAcceleration

Define

$$H_3 = \frac{a_0^2 - a_f^2}{2\bar{a}J} + \frac{v_f - v_0}{\bar{a}}, \quad (52)$$

$$H_2 = -a_f^2 + \bar{a}^2 + 2Jv_f, \quad (53)$$

$$H_0 = 3(a_f^4 - a_0^4) + 8\bar{a}(a_0^3 - a_f^3) + 24\bar{a}J(a_f v_f - a_0 v_0) \quad (54)$$

$$- 6a_0^2(\bar{a}^2 - 2Jv_0) + 6a_f^2(\bar{a}^2 - 2Jv_f) \quad (55)$$

$$+ 12J[J(v_f^2 - v_0^2 - 2\bar{a}\Delta p) - \bar{a}^2(v_f - v_0)]. \quad (56)$$

Let τ be an admissible real root of

$$\tau^4 - \frac{2\bar{a}}{J}\tau^3 + \frac{H_2}{J^2}\tau^2 + \frac{H_0}{12J^4} = 0. \quad (57)$$

Then

$$[t_1, \dots, t_7] = [(-a_0 + \bar{a})/J, H_3 - 2\tau + (J/\bar{a})\tau^2, \tau, 0, 0, 0, (a_f - \bar{a})/J + \tau]. \quad (58)$$

5.3 TerminalAcceleration

Define

$$H_3 = -\frac{a_0^2 + a_f^2}{2J\underline{a}} + \frac{\underline{a}}{J} + \frac{v_f - v_0}{\underline{a}}, \quad (59)$$

$$H_2 = a_0^2 - a_0\underline{a} + 2Jv_0, \quad (60)$$

$$H_0 = \frac{a_0^4 - a_f^4}{4} + \frac{2\underline{a}(a_f^3 - a_0^3)}{3} + \frac{\underline{a}^2(a_0^2 - a_f^2)}{2} \quad (61)$$

$$+ J[a_f^2 v_f + a_0^2 v_0 + 2\underline{a}(J\Delta p - a_0 v_0 - a_f v_f)] \quad (62)$$

$$+ \underline{a}^2(v_0 + v_f) + J(v_0^2 - v_f^2)]. \quad (63)$$

Let τ be an admissible real root of

$$0 = \tau^4 + \frac{2(2a_0 - \underline{a})}{J}\tau^3 + \frac{5a_0^2 + \underline{a}(\underline{a} - 6a_0) + 2Jv_0}{J^2}\tau^2 \quad (64)$$

$$+ \frac{2(a_0 - \underline{a})H_2}{J^3}\tau + \frac{H_0}{J^4}. \quad (65)$$

Then

$$[t_1, \dots, t_7] = [\tau, 0, (a_0 - \underline{a})/J + \tau, 0, 0, H_3 - (2a_0 + J\tau)\tau/\underline{a}, (a_f - \underline{a})/J]. \quad (66)$$

6 Fixed-time preliminary families

6.1 Constant acceleration

The zero-jerk candidate is valid when

$$j(t) = 0, \quad 0 \leq t \leq T, \quad (67)$$

$$a_f = a_0, \quad (68)$$

$$v_f = v_0 + a_0 T, \quad (69)$$

$$p_f = p_0 + v_0 T + \frac{1}{2}a_0 T^2. \quad (70)$$

6.2 Initial and terminal dwell

For a certified minimum-time profile of duration $T_{\min}$,

$$t_{\text{dwell}} = T - T_{\min}. \quad (71)$$

Initial dwell uses

$$[t_{\text{dwell}}, t_1, \dots, t_7], \quad [0, j_1, \dots, j_7], \quad (72)$$

and requires $v_0 = a_0 = 0$. Terminal dwell uses

$$[t_1, \dots, t_7, t_{\text{dwell}}], \quad [j_1, \dots, j_7, 0], \quad (73)$$

and requires $v_f = a_f = 0$.

7 Fixed-time acceleration-and-velocity families

7.1 SynchronizedAccelerationVelocity: UDDU

The directly determined phases are

$$t_1 = \frac{-a_0 + \bar{a}}{J}, \quad t_3 = \frac{\bar{a}}{J}, \quad t_5 = -\frac{a}{J}, \quad t_7 = t_5 + \frac{a_f}{J}. \quad (74)$$

Define

$$Q = \frac{1}{\bar{a}\underline{a}} \left[a_0^4 + a_f^4 - \frac{4}{3}a_0^3(2\bar{a} + \underline{a}) - \frac{4}{3}a_f^3(\bar{a} + 2\underline{a}) \right] \quad (75)$$

$$+ 2(a_0^2 - a_f^2)\bar{a}^2 + (4a_0\bar{a} - 2a_0^2)[a_f^2 - 2a_f\underline{a} + (\underline{a} - \bar{a})\underline{a} + 2J(\underline{a}T - \Delta v)] \quad (76)$$

$$+ 2a_f^2[\underline{a}^2 + 2J(\bar{a}T - \Delta v)] + 4J\{2\underline{a}[a_f\Delta v + J(-\Delta p + Tv_0)] + (\bar{a}^2 - \underline{a}^2)\Delta v + J\Delta v^2\} \quad (77)$$

$$+ 8\bar{a}J^2(\Delta p - Tv_f)] + 4a_f^2 + 2a_0^2 + (4a_f + \bar{a} - \underline{a})(\bar{a} - \underline{a}) \quad (78)$$

$$+ 4J(\underline{a} - \bar{a} + JT - 2a_f)T, \quad (79)$$

and let $q = \text{sign}(J)\sqrt{Q}$. Then

$$t_2 = \frac{-[a_f^2 - a_0^2 + 2\bar{a}^2 + \underline{a}(\underline{a} - 2\Delta a - 3\bar{a}) + 2J(\underline{a}T - \Delta v)] + \underline{a}q}{2(\bar{a} - \underline{a})J}, \quad (80)$$

$$t_4 = \frac{\underline{a} - \bar{a} + q}{2J}, \quad (81)$$

$$t_6 = T - (t_1 + t_2 + t_3 + t_4) - 2t_5 - a_f/J. \quad (82)$$

7.2 SynchronizedAlternatingAccelerationVelocity: UDUD

$$t_1 = (-a_0 + \bar{a})/J, \quad t_3 = \bar{a}/J, \quad (83)$$

$$t_5 = t_3, \quad t_7 = t_5 - a_f/J, \quad (84)$$

$$t_4 = T + \frac{-a_0^2 - a_f^2 + 2\bar{a}(a_0 + a_f - 2\bar{a}) - 2J\Delta v}{2\bar{a}J}. \quad (85)$$

Define

$$N = 3(a_0^4 + a_f^4) - 4(a_0^3 + a_f^3)\bar{a} - 4a_f^3\bar{a} + 24(a_0 + a_f)\bar{a}^3 \quad (86)$$

$$- 6(a_f^2 + a_0^2)(\bar{a}^2 - 2J\Delta v) + 6a_0^2(a_f^2 - 2a_f\bar{a} - 2\bar{a}JT) \quad (87)$$

$$- 12\bar{a}^2(2\bar{a}^2 - 2\bar{a}JT + J\Delta v) - 24a_f\bar{a}J\Delta v + 12J^2[2\bar{a}(-\Delta p + Tv_0) + \Delta v^2], \quad (88)$$

$$D = 12\bar{a}J[a_0^2 + a_f^2 - 2(a_0 + a_f)\bar{a} + 2(\bar{a}^2 - \bar{a}JT + J\Delta v)]. \quad (89)$$

Then

$$t_2 = N/D, \quad (90)$$

$$t_6 = T - (t_1 + t_2 + t_3 + t_4) - 2t_5 + a_f/J. \quad (91)$$

8 Fixed-time one-parameter root families

These families reduce the endpoint-position condition to one unknown τ . For each provisional phase vector, the engine solves

$$F(\tau) = p_7(\tau) - p_f = 0, \quad (92)$$

where p_7 is produced by sequentially applying the constant-jerk propagation equations. The implementation samples the admissible interval, brackets all sign changes, applies **fzero**, and certifies every reconstructed root.

8.1 SynchronizedTerminalAccelerationVelocity: UDDU

$$h = -\frac{(a_0^2 + a_f^2)/2 + J(-\Delta v + 2a_0\tau + J\tau^2)}{\underline{a}}, \quad (93)$$

$$[t_1, \dots, t_7] = [\tau, 0, a_0/J + \tau, T - (h - \underline{a} + a_0 + a_f)/J - 2\tau, \quad (94)$$

$$-\underline{a}/J, (h + \underline{a})/J, -\underline{a}/J + a_f/J]. \quad (95)$$

8.2 SynchronizedAlternatingTerminalAccelerationVelocity: UDUD

$$h = \frac{(a_0^2 - a_f^2)/2 + J^2\tau^2 - J(\Delta v - 2a_0\tau)}{\bar{a}}, \quad (96)$$

$$[t_1, \dots, t_7] = [\tau, 0, \tau + a_0/J, T + (h + \Delta a - \bar{a})/J - 2\tau, \quad (97)$$

$$\bar{a}/J, -(h + \bar{a})/J, \bar{a}/J - a_f/J]. \quad (98)$$

8.3 SynchronizedVelocity: UDDU

$$h = \sqrt{\frac{a_0^2 + a_f^2}{2J^2} + \frac{2a_0\tau + J\tau^2 - \Delta v}{J}}, \quad (99)$$

$$[t_1, \dots, t_7] = [\tau, 0, \tau + a_0/J, T - 2(\tau + h) - (a_0 + a_f)/J, h, 0, h + a_f/J]. \quad (100)$$

8.4 SynchronizedAlternatingVelocity: UDUD

$$h = \sqrt{\frac{a_f^2 - a_0^2}{2J^2} - \frac{(2a_0 + J\tau)\tau - \Delta v}{J}}, \quad (101)$$

$$[t_1, \dots, t_7] = [\tau, 0, \tau + a_0/J, T - 2(\tau + h) + \Delta a/J, h, 0, h - a_f/J]. \quad (102)$$

9 Root admissibility and refinement

For the minimum-time quartic families, the implementation retains roots whose imaginary part is below a scale-dependent tolerance, keeps the real part, and then applies family-specific interval restrictions. A short Newton correction is applied where the implementation provides an analytic residual derivative.

For fixed-time one-parameter families, the admissible interval is sampled at 65 points. Endpoint roots and sign-changing intervals are retained. Each bracketed interval is passed to a scalar root solver. Duplicate roots are removed after rounding to 12 decimal places.

10 Candidate certification and selection

A generated candidate is retained only when:

1. all phase durations are finite and nonnegative within tolerance;
2. phase integration reaches the terminal position, velocity, and acceleration within the implementation tolerances;
3. velocity extrema inside every constant-jerk phase respect the velocity limit;
4. acceleration respects the acceleration limit; and
5. the jerk law contains only the directed jerk magnitude, its negative, or zero.

The minimum-time engine ranks candidates by

$$T_{\text{candidate}} = \sum_i t_i. \quad (103)$$

The fixed-time engine first minimizes scalar path length and then uses integrated squared jerk as the tie breaker:

$$J_2 = \sum_i j_i^2 t_i. \quad (104)$$

11 Family index

Family	Active structure
AccelerationVelocity	Minimum-time UDDU; both acceleration bounds and the velocity bound.
TerminalAccelerationVelocity	Minimum-time UDDU; velocity and terminal-side acceleration bound.
InitialAccelerationVelocity	Minimum-time UDDU; initial-side acceleration and velocity bounds.
Velocity	Minimum-time UDDU; velocity bound with no acceleration hold.
AccelerationBounds	Minimum-time UDDU; both acceleration bounds and no velocity plateau.
Unconstrained	Minimum-time UDDU; no derivative bound is held.
InitialAcceleration	Minimum-time UDDU; initial-side acceleration bound only.
TerminalAcceleration	Minimum-time UDDU; terminal-side acceleration bound only.
Synchronized families	Fixed-time UDDU.
SynchronizedAlternating families	Fixed-time UDUD.
ConstantAcceleration	One zero-jerk phase.
InitialDwell, TerminalDwell	Zero-motion extension around a minimum-time profile.

12 References

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